

AWS re:Invent

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Advanced networking for Amazon S3

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Agenda

Introduction to virtual private cloud (VPC) endpoints

Accessing Amazon S3 using VPC endpoints

AWS PrivateLink for Amazon S3

Advanced use cases

Recap

Introduction to VPC endpoints

Introduction to VPC endpoints



Secure access to
Amazon S3 over the
Amazon network



Easy to configure,
reliable, and
highly secure



Control access with
endpoint policies

Types of VPC endpoints to access Amazon S3

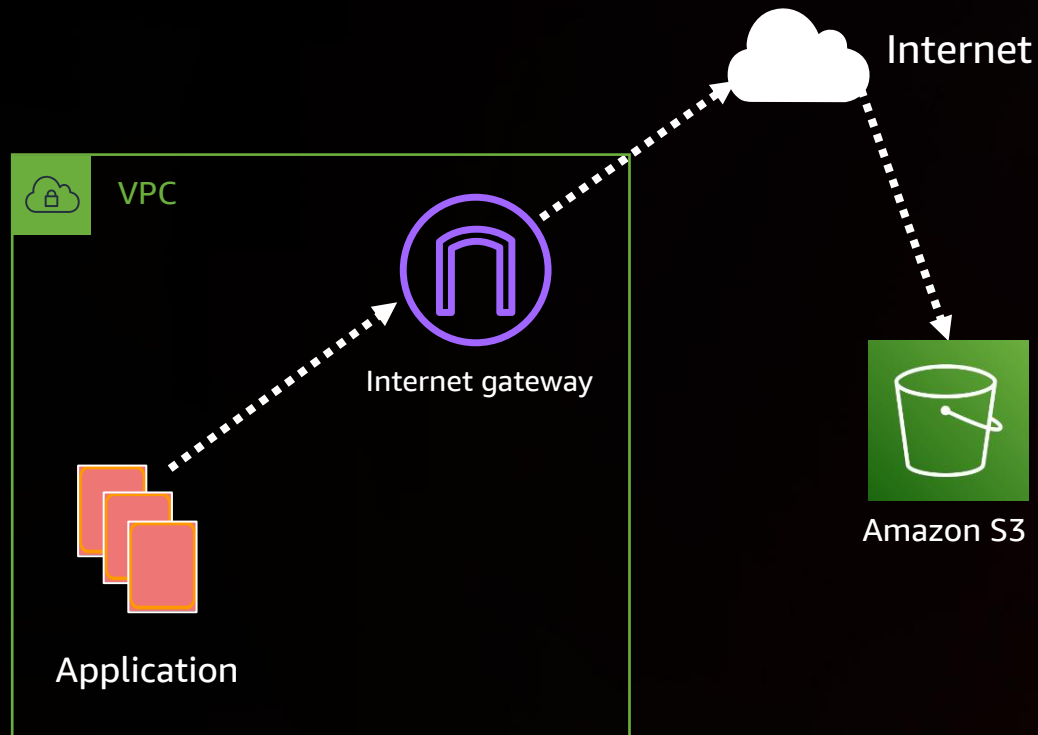
1 Gateway endpoints

2 Interface endpoints

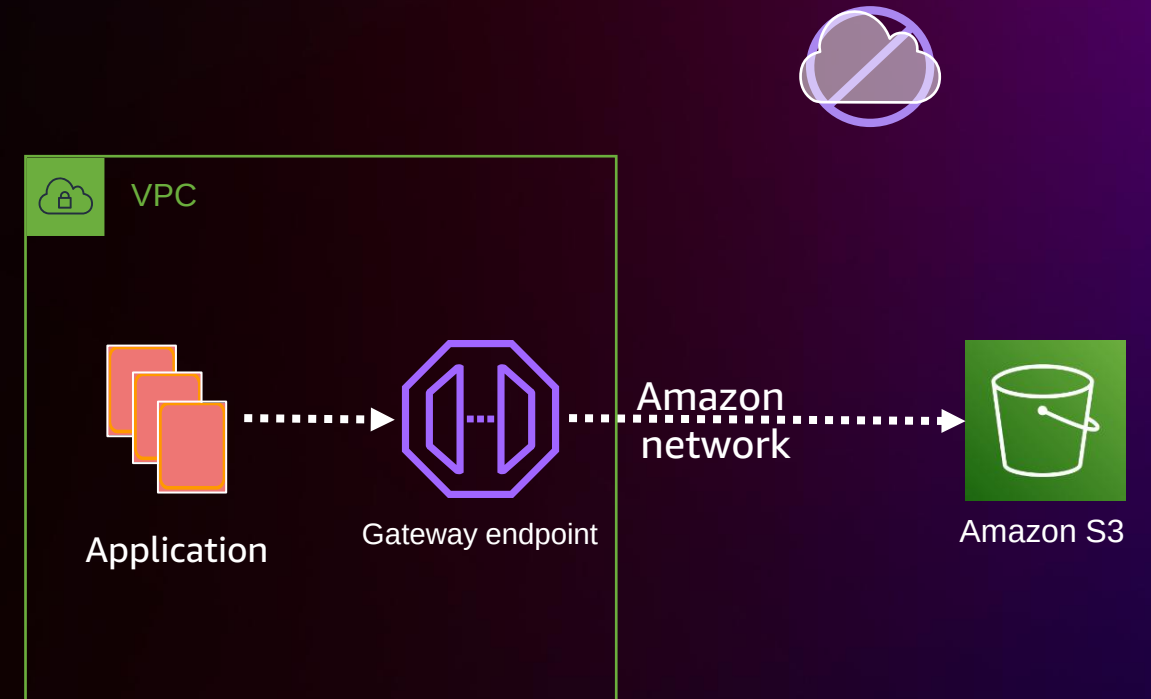
Accessing Amazon S3 using VPC endpoints

Accessing Amazon S3 from a VPC – Using gateway endpoints

Without gateway endpoint



With gateway endpoint



- Simple network architecture
- Control access via endpoint policy
- Traffic remains on Amazon network
- No additional charge to use gateway endpoints

Gateway endpoints – Based on the **routing table** in your VPC

search : s3 × Add filter

1 to 2 of 2

< > >>

	Service Name	Owner	Type
☒	com.amazonaws.us-east-1.s3	amazon	Gateway
☐	com.amazonaws.us-east-1.s3	amazon	Interface

VPC* vpc-c8b3a8b2 ▼ ↻ ℹ

Configure route tables

A rule with destination **pl-63a5400a (com.amazonaws.us-east-1.s3)** and a target with this endpoints' ID (e.g. vpce-12345678) will be added to the route tables you select below.

Subnets associated with selected route tables will be able to access this endpoint.

rtb-7cc3a502 ×

	Route Table ID	Main	Associated With
☒	rtb-7cc3a502	Yes	6 subnets

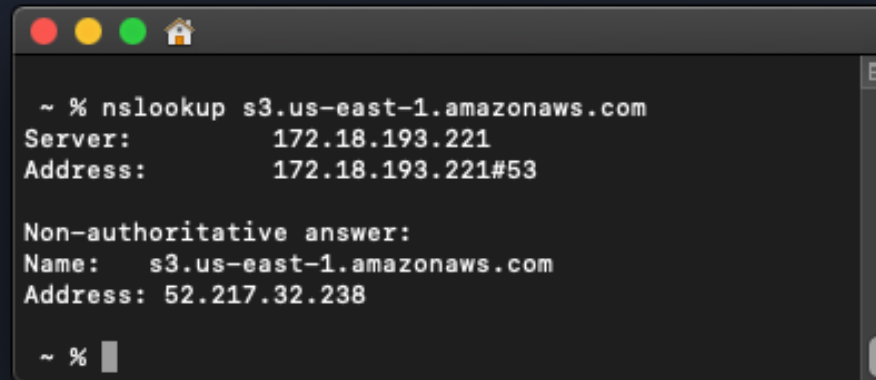
Gateway endpoints – Use standard S3 DNS

Standard Amazon S3 Regional DNS names

Ex: `s3.us-east-1.amazonaws.com`



Amazon S3
public IPs

A terminal window with a dark background and light text. It shows the output of an nslookup command for the domain s3.us-east-1.amazonaws.com. The output includes the server address (172.18.193.221) and a non-authoritative answer with the IP address 52.217.32.238.

```
~ % nslookup s3.us-east-1.amazonaws.com
Server:      172.18.193.221
Address:     172.18.193.221#53

Non-authoritative answer:
Name:   s3.us-east-1.amazonaws.com
Address: 52.217.32.238

~ %
```

AWS PrivateLink for Amazon S3



AWS PrivateLink for Amazon S3

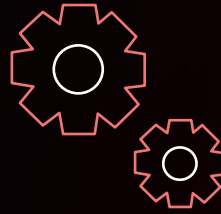


Data centers



Private

Traffic traverses
on my and
Amazon network
rather than the
internet



Simple

Simple configuration
and reliable
performance over
AWS Direct Connect



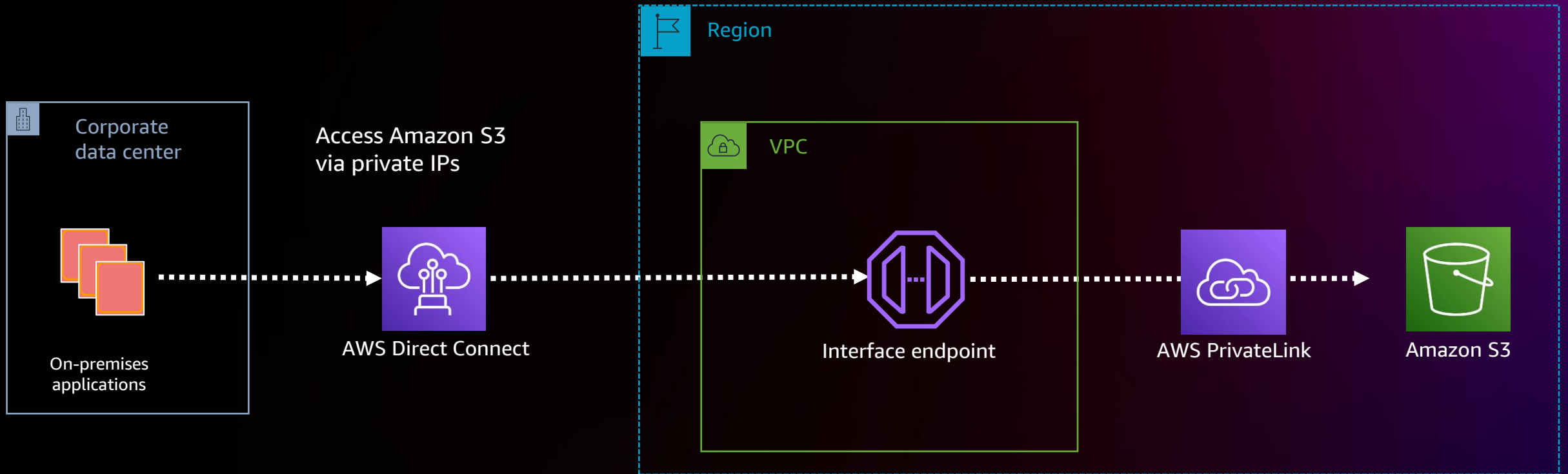
Performance

Avoid internet
congestion and use
Direct Connect for
better performance



AWS PrivateLink

Accessing S3 from a VPC – Using interface endpoints



Interface endpoints – Based on the **private IPs** in your VPC

The screenshot displays the AWS Management Console interface. At the top, a search bar contains 's3' and a filter button. Below it, a table lists service endpoints:

Service Name	Owner	Type
☐ com.amazonaws.us-east-1.s3	amazon	Gateway
☒ com.amazonaws.us-east-1.s3	amazon	Interface

Below the table, the VPC is identified as 'vpc-c8b3a8b2'. A red box highlights the 'Subnets' section, which shows a list of subnets with their availability zones and IDs:

Availability Zone	Subnet ID
☒ us-east-1a (use1-az4)	subnet-4baf3106
☒ us-east-1b (use1-az6)	subnet-68ea2137
☒ us-east-1c (use1-az1)	subnet-7b7fb01d
☒ us-east-1d (use1-az2)	subnet-7503d654
☒ us-east-1e (use1-az3)	subnet-c2ffedfc
☒ us-east-1f (use1-az5)	subnet-ef8505e1

Interface endpoints – Use endpoint-specific DNS

Regional endpoint-specific DNS names

`vpce-1a2b3c4d-5e6f.s3.us-east-1.vpce.amazonaws.com`



Private IPs of ENI in subnet 1

.
.

Private IPs of ENI in subnet 6

Zonal endpoint-specific DNS names

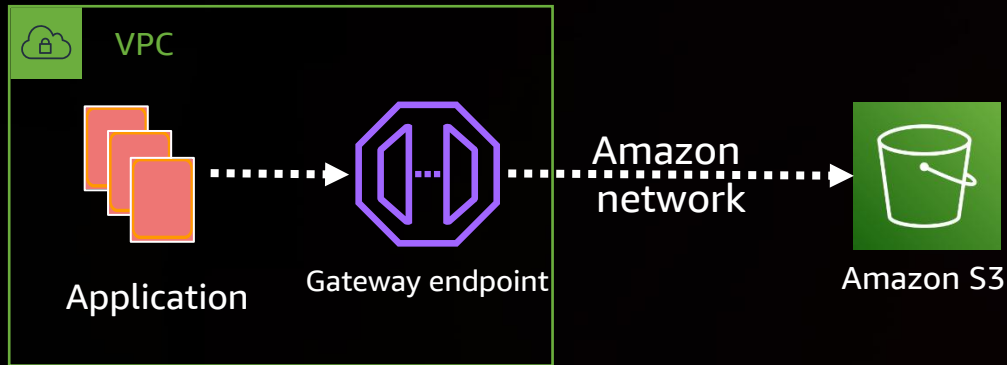
`vpce-1a2b3c4d-5e6f-us-east-1f.s3.us-east-1.vpce.amazonaws.com`



Private IP of ENI in subnet 1

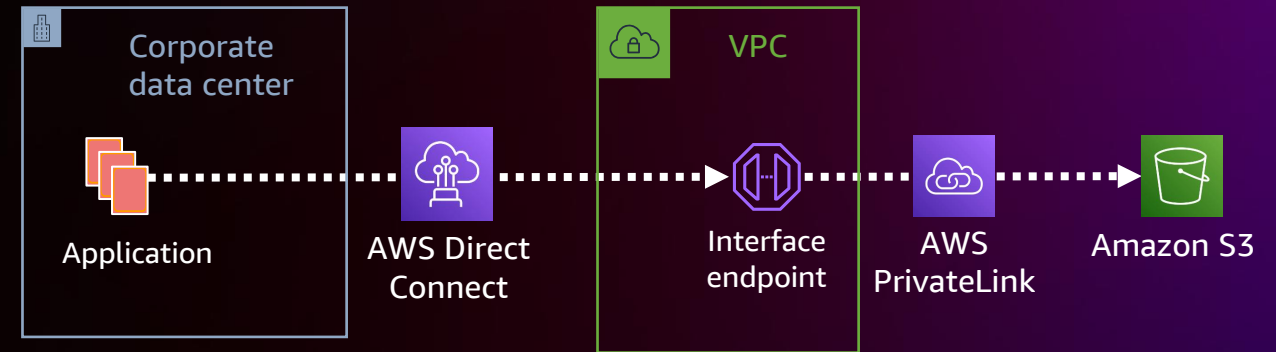
VPC endpoints – Gateway and interface

Gateway endpoints



- Clients use public S3 IP addresses
- Routes based on routing table rule with target gateway endpoint
- No support for on-premises applications or cross-Region requests
- No additional charge

Interface endpoints Based on AWS PrivateLink



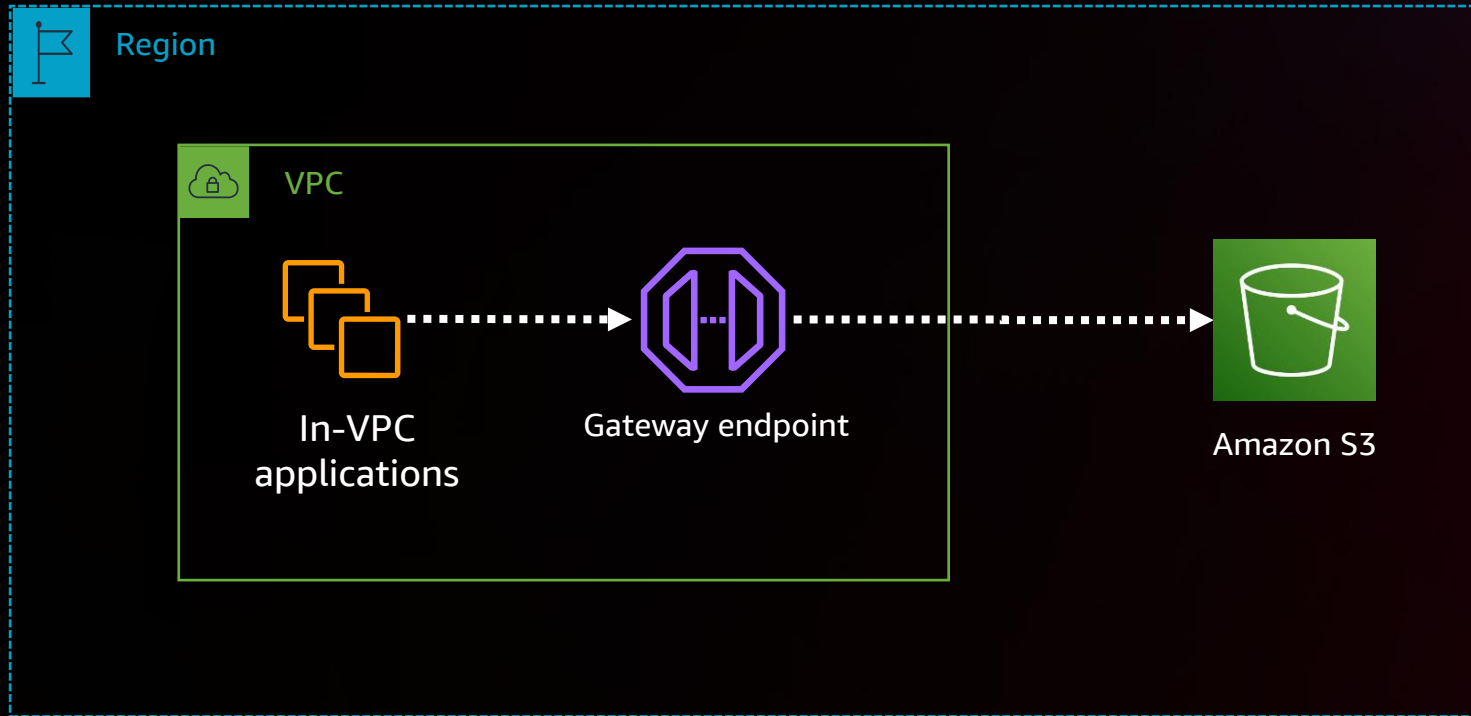
- Clients use IP addresses from your VPC
- Routes based on DNS entries for your Regional and ENI level
- Support for on-premises applications and cross-Region requests
- Paid feature

Advanced use cases

Access Amazon S3 from:

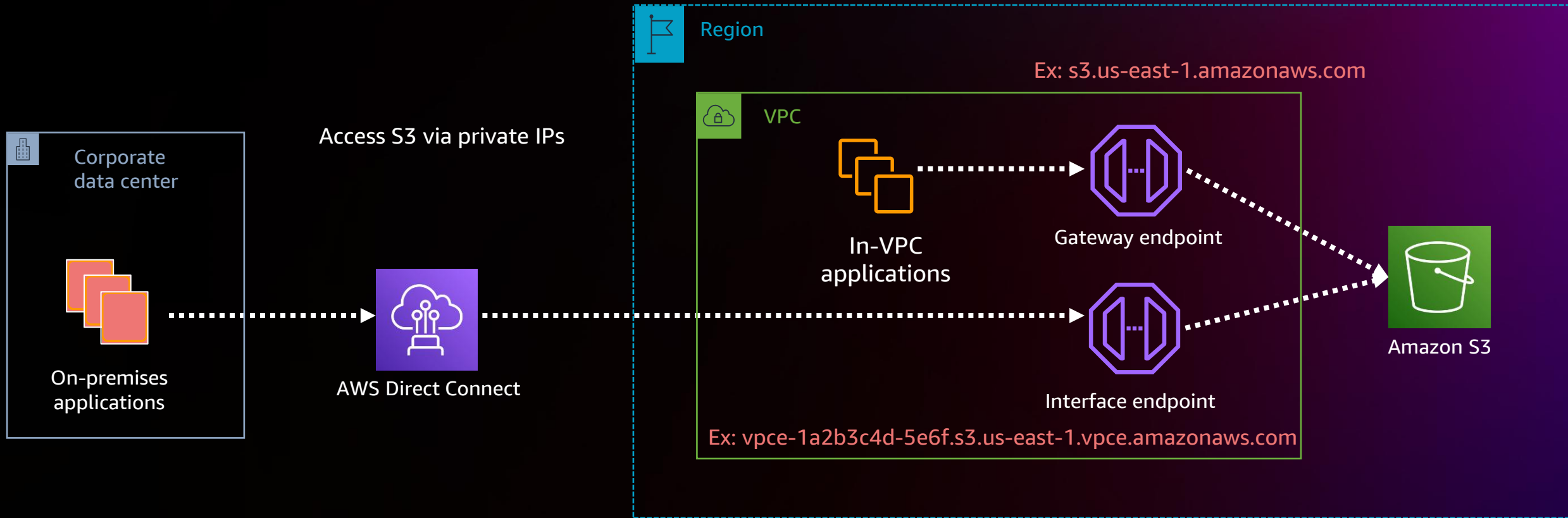
- VPC
- On-premises
- On-premises and multiple VPCs
- Cross-Region VPC
- VPC using a Multi-Region Access Point

Case 1: Access Amazon S3 from VPC



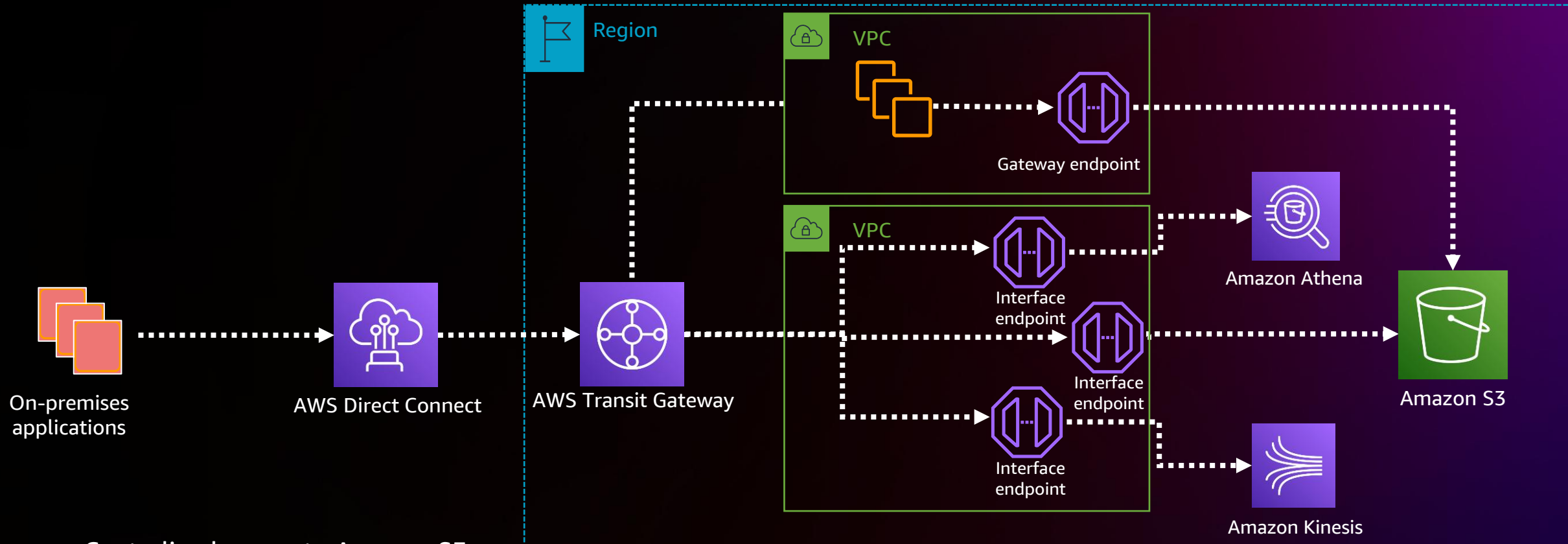
Amazon S3 gateway endpoints provide private access to Amazon S3 for VPC resources

Case 2: Access Amazon S3 from **on-premises**



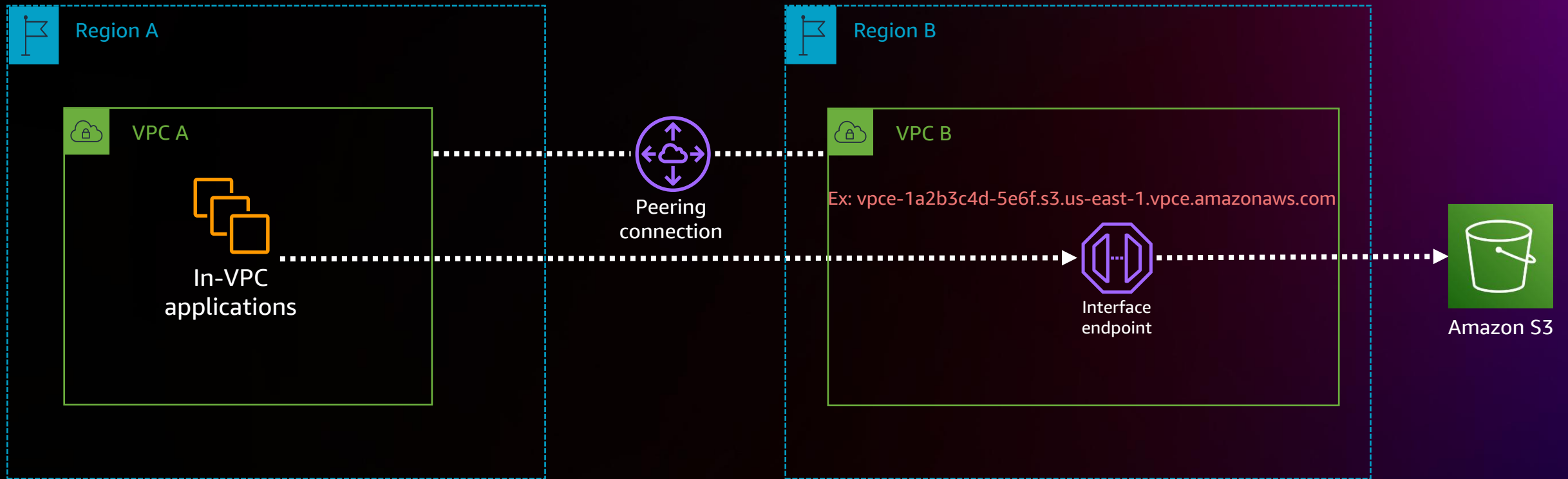
- **Optimize** cost
- **Leverage** existing VPC configuration with Direct Connect

Case 3: Access Amazon S3 from **on-premises** and **multiple VPCs**

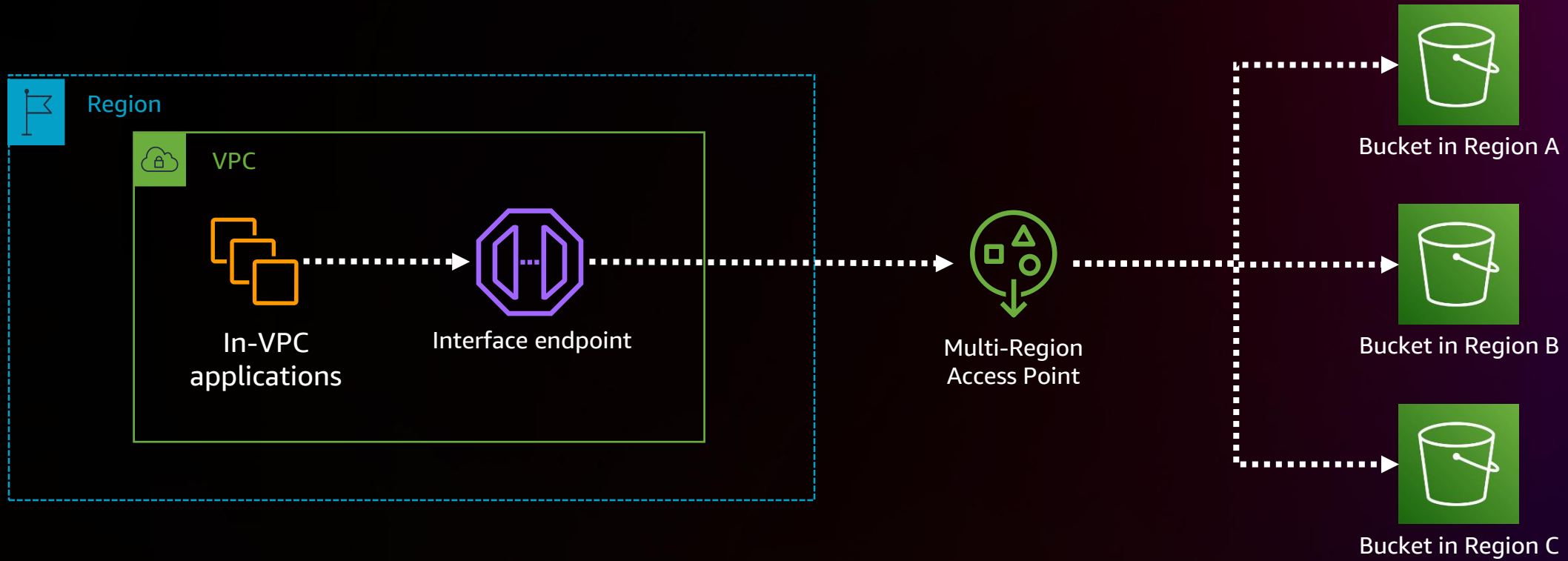


- Centralized access to Amazon S3
- Simplify VPC endpoint policy management

Case 4: Accessing Amazon S3 from cross-Region VPC



Case 5: Accessing S3 using Multi-Region Access Point from **VPC**



Recap

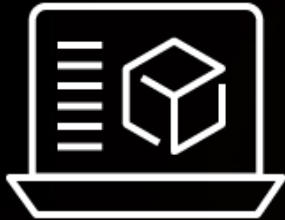


Recap

- VPC endpoints are endpoints in a VPC that provide secure access to Amazon S3 over the Amazon network; you do not need to configure an internet gateway or NAT devices
- Use gateway endpoints to connect from within VPC to Amazon S3, and use interface endpoints to connect from on-premises to Amazon S3
- AWS PrivateLink for Amazon S3 provides interface endpoints for Amazon S3 to give you private access from on-premises in a simple configuration
- Use cases in which you should use gateway endpoints, interface endpoints, and both endpoints together while connecting from in-VPC, on premises, and VPC-VPC
- Amazon S3 from VPC, on-premises, multiple VPCs, cross-Region application, and use of Multi-Region Access Points

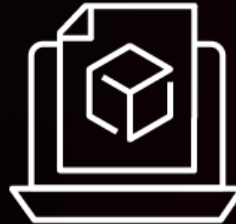
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Thank you!

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survey in the **mobile app**

